Vascular Access

23

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23.1 Introduction

The execution of the complex procedure of hematopoietic cell transplantation (HCT) requires a stable and safe venous access. This aim is usually achieved by the insertion of a central venous catheter (CVC). CVC is a key tool to ensure a safe central venous access for multiple purposes (chemotherapy, parenteral nutrition, supportive therapy, hydration, and blood sampling), to reduce the need for repeated, painful venipunctures, and to preserve the peripheral venous asset. The use of CVC is associated with the occurrence of complications (malfunctioning, occlusion, dislodgement, kinking, rupture, thrombosis, and catheter-related bloodstream infections-CRBSI) that can cause its premature removal or be life-threatening for the patient. For this reason, education and continuous training of health personnel are fundamental to preserve the CVC life span from complications. Table 23.1 shows the essential steps of CVC maintenance (Cesaro et al. 2016; Cellini et al. 2022).

CVCs are being designated by:

- Duration (e.g., temporary or short-term versus permanent or long-term).
- Site of insertion (e.g., subclavian vein, femoral vein, jugular vein, and basilic vein).
- Number of lumens (single, double, or triple lumen).
- Characteristic of tip (open tip or closed tip).
- Materials to reduce complications (e.g., impregnation with heparin, antibiotics, or silver).

23.2 Type of CVC Materials

Catheter materials should be biocompatible, kink resistant, inherently chemically resistant, and neutral, biostable, soft, and deformable and should have a high tensile strength (Lim et al. 2018; Crocoli et al. 2022). The most commonly used materials are polyurethane and silicone. Silicone catheters are flexible, chemically stable, and well tolerated. Polyurethane has a superior tensile strength. Non-tunneled, semirigid catheters are usually made of polyurethane, while tunneled catheters are usually made of both silicone and polyurethane (Lim et al. 2018). The choice of a polyurethane or a silicone catheter is debated. Silicone catheters are more prone to material failure as a result of the development of surface irregularities due to loss of barium sulfate molecules and thrombotic occlusion. The introduction

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Table 23.1 CVC maintenance: recommended rules

1. Assessment of line functionality and dressing site daily for inpatients or every 2–3 days for outpatients
2. CVC care and maintenance as dictated by local policy or standard operating procedure[a]
3. Vigorous mechanical scrub for manual disinfection prior to each CVC access and allow it to dry. Acceptable disinfecting agents include 70% isopropyl alcohol, iodophors (i.e., povidone-iodine), or >0.5% chlorhexidine in alcohol solution
4. Change gauze dressing every 7 days or before in case of soiled, dampened, and loosened dressing
5. Change the transfusion administration set and filter after the completion of each unit or every 4 h. If more than 1 unit can be infused in 4 h, the transfusion set can be used for a 4-hour period
6. Change intermittent administration sets every 24 h
7. Replace administration sets for parenteral nutrition solutions at least every 24 h
8. Replace administration sets used for intravenous fat emulsions infused separately every 12 h
9. Change caps every 72 h (or 7 days if pressure-positive or neutral needle free connector is used)

[a]There may be a variability among EBMT centers regarding the practice of CVC care and maintenance such as the use of sterile gloves and mask by provider/assistant, the adoption of aseptic technique for all catheter entries, the use of prepackaged dressing change kit, the frequency of flushing, and the type of solution used for flushing CVC

of third-generation polyurethane catheters with power-injectable technology, that allows flows up to 5 mL/s and contrast medium injection, made this type of CVC a favorite with many operators (Crocoli et al. 2022).

23.3 Type of CVC

CVCs are classified in two main categories: *tunneled and non-tunneled*, according to whether or not they follow a subcutaneous route before accessing the central vein. Non-tunneled catheters are directly inserted into a peripheral or large central vein. Both tunneled and non-tunneled CVCs may have a single or multiple lumen. Tunnelling of CVCs was introduced to reduce the risk of infectious and mechanical (dislodgement) complications, and this type of CVC is ideal for long-term care (Cellini et al. 2022). Non-tunneled CVCs are usually inserted for a short to medium period (from 2–4 weeks to 1–3 months) (Lee and Ramaswamy 2018; Padmanabhan 2018). Tunneled CVCs are in turn classifiable in two subgroups: partially implanted and totally implanted. Partially implanted CVCs are characterized by an external part outside the patient's body whose extremity (hub) is used to draw blood sampling or to connect the infusion lines, a tunneled subcutaneous part with a Dacron cuff at a few centimeters from the skin entry point, and a final intravenous part with the tip positioned at the border between the superior vena cava and the right atrium. The Dacron cuff stimulates a fibrotic reaction of the subcutaneous tissues over 2–4 weeks ensuring stability and CVC securement. Both cuff and subcutaneous course are fundamental to prevent the CVC from becoming infected due to the migration of external microbes along the CVC.

The tip of CVC can be open or closed: open tip requires the clamping of the external part of CVC when it is not in use to avoid blood backflow with breath or body movements, while closed tip has lateral valves that open as fluid is withdrawn or infused and remain closed when the CVC is not used. The clamping of CVC with open tip is not needed if a neutral pressure needle-free connector is used to prevent blood backflow.

Totally implanted catheters consist of a reservoir (port) placed in a pocket in the subcutaneous tissue, anteriorly on the chest wall, below the clavicle, that is connected to the catheter. This type of CVC preserves the patient's body image and ensuring almost no limitations on sports activities, and body hygiene. Accessing to this type of CVC needs skin puncture with a special "non-coring" needle (Huber needle or gripper system). In case of frequently accesses, the procedure can be painful or discomforting for the patient, requiring the application of topical skin anesthetic for its prevention. "Non coring" needle does not permit the infusion or the extraction of high volumes making it less suitable for patients requiring high infusion or blood extrac-

tion rates. The recent introduction of port models with a modified reservoir chamber (vortex, tidal, power port) has allowed to obtain a higher flow rate suitable for leukapheresis, red blood cell exchange, extracorporeal photopheresis, and therapeutic plasma exchange (Blanco-Guzman 2018; Lim et al. 2018).

The *peripherally inserted central catheter (PICC)* is a CVC inserted into a vein of the arm, usually the basilic or cephalic veins; its tip is advanced through the axillary and subclavian veins up to the cavo-atrial junction (Cornillon et al. 2017). For more information on PICCs, see Chap 32.

23.4 Venous Access

Central lines are usually inserted through the subclavian, the jugular, or, less frequently, the femoral vein. This last venous access is associated with a higher risk of infectious complications (O'Leary 2016), and it is more commonly used in critically ill patients admitted to intensive care units who require a non-tunneled CVC. It is recommended that the ratio of catheter caliber to vein diameter should not exceed 1/3. Using the subclavian or jugular access, the tip of the catheter has to lie in the superior vena cava, just before the entrance of the right atrium, about 29–55 mm below the level of trachea carina (in adults). The incidence of pneumothorax after CVC insertion is about 1.5–3.1%, and it is higher with subclavian vein catheterization, whereas the risk of hemorrhage and bruise is slightly more common with the jugular venous line access.

In the positioning of a PICC, the incannulation of the basilic vein is preferred to that of the cephalic vein as it has low risk of complications. To minimize the risk of complications due to venous catheterization, the routine use of ultrasound guidance to cannulate the vein is recommended instead of the classical (blind) technique (Crocoli et al. 2015; Crocoli et al. 2022).

A chest X-ray must be performed at the end of the CVC insertion procedure to confirm that the line is positioned inside the superior vena cava, and for the cannulation of subclavian or jugular veins, no pneumothorax was inadvertently caused. Alternatively, the use of intracavitary ECG (electrocardiographic method) is a noninvasive method to evaluate the correct position of the catheter tip.

23.5 CVC Complications

Catheter-related complications may be classified into *infectious* (local or systemic) and *mechanical* (occlusion, rupture, dislodgement, accidental self-removal, and thrombosis) (Cesaro et al. 2009). As the catheter is itself a risk for developing complications, it should be removed as there is no further need for it. Premature CVC removal is indicated in case of mechanical complication with persistent malfunction, CRBSI by *Candida* spp., *Pseudomonas* spp., *Klebsiella* spp., *Staphylococcus aureus,* persistent bacteria colonization, recurrent CRBSI, or CVC thrombosis.

23.5.1 Special Measures to Prevent Catheter-Related Infections

The key rules to prevent infections are proper handwashing by the performing provider, the use of aseptic technique over the patient at insertion time, thorough cleaning of the insertion site, and periodic review of the CVC exit site (Cesaro et al. 2016; Cellini et al. 2022). To prevent CRBSI and tunnel or exit-site infection, medication-impregnated dressings with different antimicrobial materials were developed to decrease the production of the biofilm by microorganisms and decrease the adhesion of them to the catheter walls. Impregnating medications may contain chlorhexidine gluconate, silver sulfadiazine, rifampin, and minocycline. Chlorhexidine gluconate is among the most used, by impregnating the whole dressing or applying an impregnated sponge (e.g., Biopatch®) and covered by a transparent polyurethane semipermeable transparent dressing. The efficacy of flushing or locking CVC with an antibiotic solution to prevent infec-

tion is debated, and it is associated with the risk of increasing microbial antibiotic resistance; therefore, it is not recommended and it should be reserved to selected cases (van den Bosch et al. 2021).

23.6 Catheters for Leukapheresis

The procedure of stem cell collection by apheresis is performed both for auto- and allo-HCT to obtain stem cells (O'Leary 2016). As the procedure requires sustained high blood flow rates (50–100 mL/min), an adequate venous access is needed. Peripheral access placed in the basilic, cephalic, brachial, median cubital, and radial veins is recommended. Considering that the placement of a central CVC is associated with potentially life-threatening complications such as pneumothorax, bleeding, and embolism, its use is not recommended for PBSC collection of a healthy volunteer donor. Conversely, in the case of auto-PBSC, if the patient has no adequate peripheral or central venous access, a temporary non-tunneled CVC may be placed in the internal jugular, subclavian, or femoral veins. Catheter removal is performed on donor laboratory values (platelets >50 × 10^9/L) and after the assessment of an adequate $CD34^+$ dose collection and cryopreservation.

Partially implanted silicone CVCs are often used by pediatric oncologists-hematologists because they are most suitable for long-term complex treatment. In the case of leukapheresis procedure, silicone CVCs are not ideal because they are more prone to collapse during automatic apheresis (Ridyard et al. 2017). On the other hand, the harvesting procedure of peripheral stem cell collection, which requires high blood flow rates and a large needle, may be difficult in children below 10 kg using a temporary peripheral venous access due to the small size of veins (Padmanabhan 2018). In this case, the placement of a short-term CVC made of polyurethane may be needed (Cooling 2017). In younger children, the rigidity of such material and the narrower lumens of the veins may represent a potential risk for thrombosis and infection.

Key Points

CVC indications and insertion		
1. Type of CVC	Tunneled CVCs/port/PICCs	Long-term therapy (months, years)
		Port for intermittent use, tunneled CVC for continuous use
		Suitable for inpatient and outpatient
	Non-tunneled CVCs	Short-term therapy (2–4 weeks, 1–3 months)
		Suitable for inpatient
2. Number of lumens	Single lumen vs double lumen	Double lumen in patients undergoing HCT, critically ill patients, intensive intravenous therapy
3. Insertion	Percutaneous/minimally invasive	Ultrasound guidance recommended
		Adequate training required
	Cutdown approach	Very limited indication (premature infants)
		Experienced operator
4. Material	Silicone	Tunneled CVC
	Polyurethane	Tunneled and non-tunneled CVC

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